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Envoix: A Hybrid Adaptive Platform for Secure End-to-End File Transfer

VE441 · App Development for Entrepreneurs · Project Proposal

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Envoix — Elevator Pitch

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Never waste time choosing how to send a file again —
Envoix automatically selects the fastest, most secure transfer path across any platform and any network.

- **Zero decision cost** — routing is automatic, zero manual selection
- **Uniform E2EE** — all transfer modes, all files, always encrypted
- **Encrypted chat** — conversation context stays as private as the file
- **No account required** — pair instantly via QR code or short token

Outline

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Team Members

Team: NUUB **PM:** Zhang Jinbin

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Name	jAccount	Task Assignment	Technical Strengths
Sun Qizhen	qz_sun	Product architecture, routing engine, network path scoring	Networking, distributed systems, system design
Zhang Jinbin	moranxuege	PM, mobile client (Android/iOS), cross-platform UI/UX	Android (Kotlin), iOS (Swift), UX implementation
Li Yuhao	17521218310	End-to-end encryption, QR/-token handshake, encrypted chat	Cryptography, backend security, protocol design
He Yumeng	y4954he	Relay server, deployment, testing	Backend, QA, Rust REST

Affinity Map

chkxw.github.io/affinity-map-site

Customer Profile — Who Are Our Users?

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User Context (17 interviews)

- Own **2–3 devices**: phone + laptop (± tablet)
- Platforms span **iOS, Android, Windows, macOS, Linux**
- **8 / 17** transfers are self-to-self between own devices
- Files: art / photos, coursework docs, large datasets

Current Tools

WeChat File Transfer	universal default (10 / 17)
Google Drive	large files
Discord / QQ	makeshift shuttles
USB cable / drive	offline fallback

Key Insight

Tool choice is driven by **habit, not quality**.

"WeChat was already open, so I just used it. I wasn't thinking about AirDrop."

Even full Apple-ecosystem users prefer WeChat over AirDrop due to inertia.

What They Like Today

- **Always there** — no extra install needed
- **Persistent storage** — QQ files don't expire
- **Multi-purpose** — chat + transfer in one app

Customer Profile — Key Pain Points

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Pain Point	Representative Quote
Transfer interrupted; no auto-resume ($\times$ 11)	*"scp cut out halfway and I had to start over from the beginning"*
Cross-platform wall: multi-step detour ($\times$ 7)	*"I had to log in on both devices, sync through WeChat, download, then move it"*
File size & format caps ($\times$ 7)	*"Discord has a file size limit; for anything larger I use Google Drive"*
Cloud storage quota exhaustion ($\times$ 6)	*"I've already used 3 Google accounts worth of the free 15 GB"*
Speed throttling behind paywall ($\times$ 6)	*"Cloud storage requires a paid membership for full speed"*
WeChat save-location friction ($\times$ 4)	*"WeChat automatically saves files somewhere weird; you have to Save As every time"*

$\times$ = affinity map tally; all findings supported by $\geq$2 interviewees.

Value Proposition

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For multi-device users who lose files to interruptions and platform walls, ***Envoix*** *automatically routes every transfer via the fastest available path — LAN, P2P, or relay — with uniform end-to-end encryption, no account, no size cap, and no speed throttle.*

Pain Point (×)	**How Envoix Addresses It**
Interrupted; no auto-resume (11)	Built-in resumable transfer; reconnects silently after any network drop
Cross-platform wall (7)	Runs on iOS, Android, Windows, macOS, Linux — one app, zero friction
File size & format caps (7)	No server storage; size bounded only by local disk
Storage quota & speed throttle (6)	LAN direct path at full NIC speed; relay is ciphertext-only fallback
Privacy resignation (6)	E2EE before leaving the device; relay never sees plaintext

Storyboard — "Lin's 11 PM Deadline"

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1 · The Deadline Lin is finishing her design project at 11 PM. She needs to send a 2 GB video render to her Windows teammate before midnight. ⚠ *Rushed, under pressure*	**2 · Three Tools Fail** AirDrop: "Apple only." Discord: file too large. WeChat: starts, then drops with no resume. 15 minutes wasted. ☹ *Frustrated, helpless*	**3 · Opens Envoix** Teammate suggests Envoix. She opens it — no sign-up screen. "Send File" is the first thing she sees. ☻ *Cautiously hopeful*
4 · One Scan to Pair Teammate shows a QR code on his screen. She scans it. Devices paired in under 2 seconds. She taps the file and hits Send. *Surprised by simplicity*	**5 · LAN Direct Transfer** Envoix detects both are on the same Wi-Fi. Routes directly over LAN — no cloud, no relay. 2 GB transferred in 40 seconds at full speed. ★ *Pleasantly astonished*	**6 · File Received, Deadline Met** File arrives intact, original quality. "That was it?" Project submitted on time — no cloud, no drama. ♥ *Relieved and delighted*

Pain points

resolved: interruption (×11) · cross-platform wall (×7) · file size cap (×7)

Market Overview

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The Fundamental Gap

Existing systems typically optimize for **exactly one transport pathway**. No single consumer app combines all approaches into a unified, adaptive experience.

Cloud Storage
Google Drive
iCloud · Dropbox

Messaging Apps
WeChat
WhatsApp ·
Telegram

LAN
AirDrop
Nearby Share

Internet P2P
BitTorrent
Syncthing

PAKE Transfer
Magic Wormhole
croc

Competitors analyzed: Google Drive · WeChat · AirDrop · BitTorrent · croc

Competitor Profiles (1/2)

Google Drive — Cloud Storage

- \+ Persistent storage
- \+ Cross-device sync
- \+ 15 GB free storage
- \+ No recipient installation
- \+ Fast transfer
- − Double bandwidth usage
- − Metadata leakage
- − Subscription pressure
- − Poor for transient transfer
- − Privacy concerns

WeChat — Messaging App

- \+ 1.3B+ users
- \+ Already widely used
- \+ Fast and convenient
- − 1 GB file limit
- − Aggressive compression
- − Multiple local copies
- − No persistent storage
- − Ecosystem lock-in
- − Surveillance/privacy risks

Competitor Profiles (2/2)

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AirDrop — LAN

- + Bluetooth + Wi-Fi Direct
- + No account required
- + No internet required
- + Seamless Apple UX
- − Apple-only
- − Proximity-only
- − No remote fallback
- − Discovery may leak info

BitTorrent — Internet P2P

- + Cross-platform
- + Widely used
- + Fast P2P transfer
- + Secure transfer
- − Optimized for many peers
- − Weak in small networks

croc — PAKE Transfer

- + SPAKE2 cryptographic model
- + Relay stores no plaintext
- + Resistant to offline attack
- + Open-source
- + Closest arch to Envoix
- − Desktop CLI-only
- − No LAN fast path
- − All traffic via relay

Comparison Table

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Feature	Google Drive	WeChat	AirDrop	BitTorrent	croc	Envoix
Cross-platform	✓	Partial	×	✓	✓	✓
LAN / local transfer	×	×	✓	×	×	✓
Internet / relay	✓	✓	×	×	✓	✓
P2P transfer	×	×	✓	✓	×	✓
Auto-routing	×	×	×	×	×	✓
Adaptive LAN + relay	×	×	×	×	×	✓
E2EE (all modes)	×	×	Partial	Partial	✓	✓
No account required	✓	×	✓	✓	✓	✓
No ecosystem lock-in	✓	×	×	✓	✓	✓
Mobile app	✓	✓	✓	Partial	×	✓
Resume on failure	Partial	×	×	Partial	×	✓
Open-source	×	×	×	✓	✓	✓

Part I

Appendix

Interview Design — Participants & Venue

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How We Find Interviewees

- SJTU engineering/CS communities
- GitHub discussions, V2EX, WeChat tech groups
- Screening: regularly transfer files across multiple devices/OS, and have encountered at least one of: size limits, incompatibility, network failure, privacy concerns
- **Avoid friends/family** — reduce social desirability bias

Venue & Session Structure

- Remote via Tencent Meeting / Zoom
- 10–20 min per session
- One interviewer + one note-taker
- Recorded with consent

0–5 min	Warm-up, device context
5–15 min	Pain stories, failure scenarios
15–20 min	Ideal workflow, retention

Theme 1: Device Context & Current Behavior

Goal: Establish devices/platforms used and current tools, without leading toward any product concept.

Q1 Walk me through the devices you use daily and how often they need to share files with each other.

Q2 Think back to the last few weeks — what triggered your need to send a file, and to whom?

Q3 What apps or methods do you normally reach for? How did you settle on those tools?

Q4 When you're about to send a file, what goes through your head before deciding which tool to use?

Theme 2: Pain Points & Failure Stories

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Goal: Surface real breakdown moments, workarounds, and emotional friction.

Q5 Walk me through the last time you sent a large file — from the moment you decided to send it to delivery.

Q6 Has your usual method ever just stopped working? What happened, and what did you do instead?

Q7 When transferring between iPhone and Android, or phone and a different-OS computer, how do you handle that?

Q8 Describe a time when security/privacy crossed your mind during a file transfer. Did it change your behavior?

Q9 What does it feel like when a transfer fails or arrives corrupted? Tell me about a specific moment.

Theme 3: Motivations, Ideals & Accepted Friction

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Goal: Understand retention drivers, unarticulated needs, and normalized workarounds.

Q10 When a file transfer tool works really well, what makes you keep coming back to it?

Q11 Describe your ideal file-sending experience for a scenario that comes up regularly for you.

Q12 What frustrations have you accepted as “the way file transfer works” — things you’ve stopped trying to fix?

Thank You